

Data Availability Statement

Gravitational Wave Data

The gravitational wave data used in this study are publicly available through the Gravitational Wave Open Science Center (GWOSC) at <https://www.gw-openscience.org/>. Specifically, we used:

- GW150914 strain data: <https://www.gw-openscience.org/events/GW150914/>
- GW190521 strain data: <https://www.gw-openscience.org/events/GW190521/>

The raw data are provided in HDF5 format, and we used the 16 kHz downsampled data for our analysis.

Analysis Products

All analysis products generated in this study, including: - Processed waveform data - Parameter estimation posterior samples - Statistical significance estimates - Spectral analysis results

will be made available upon publication of this manuscript at <https://zenodo.org/record/XXXXXXX> (DOI to be generated upon acceptance).

Analysis Code

The custom analysis code developed for this study will be publicly released as open-source software upon acceptance of this manuscript. The code repository will be available at <https://github.com/author/subharmonic-gw-analysis> and will include:

- Core analysis modules
- Waveform generation routines
- Parameter estimation pipelines
- Plotting and visualization tools
- Documentation and examples
- Configuration files used for results in the paper

Reproducing Results

To facilitate reproducibility, we provide a Docker container with all necessary dependencies pre-installed. The container can be accessed at <https://hub.docker.com/r/author/subharmonic-gw-analysis> upon publication.

Instructions for reproducing the key results are available in the README file of the code repository. All figures in the paper can be regenerated using the provided scripts.

Long-Term Archiving

For long-term archiving, all data products and code are additionally preserved in the institutional repository of AELF Research Archive at <https://archive.aelf.io/collections/PHY-CQG-011>, which guarantees a minimum 10-year retention period.

Access Restrictions

There are no restrictions on the use of the data and code provided that proper attribution is given through citation of this paper and original data sources.